



Name: _____

Date: _____

1. Factorize $x^2 - 6x - 7$

A. $(x + 1)(x + 7)$

B. $(x + 1)(x - 7)$

C. $(x - 1)(x + 7)$

D. $(x - 1)(x - 7)$ ☐

2. Factorize $49p^2 + 9q^2 - 42pq$

A. $(7p + 3q)(7p - 3q)$

B. $(7p + 3q)^2$

C. $(7p - 3q)^2$

D. $(49p + 9q)^2$ ☐

3. Which of the following do(es) not have $x + 2$ as a factor?

I. $x^2 + 4$

II. $x^2 - 4$

III. $(x - 3)^2 - 25$

A. I only

B. II only

C. I and III only

D. II and III only ☐

4. Factorize $x^2 - 8xy + 15y^2 - 5x + 15y$

A. $(x - 3y)(x - 5y - 5)$

B. $(x - 5y)(x - 3y - 5)$

C. $(x + 3y)(x - 5y - 5)$

D. $(x + 5y)(x - 3y - 5)$ ☐

5. Factorize $x^2(x + y) - y^2(y + x)$

A. $(x + y)^2(x - y)$

B. $(x + y)(x^2 + xy + y^2)$

C. $(x - y)^2(x + y)$

D. $(x - y)(x^2 - xy + y^2)$ ☐

6. Factorize $y^2 + 4y - 12$

A. $(y + 2)(y - 6)$

B. $(y + 3)(y - 4)$

C. $(y - 2)(y + 6)$

D. $(y - 3)(y + 4)$ ☐

7. Factorize $n^2 + 12n + 35$

A. $(n + 5)(n + 7)$

B. $(n + 5)(n - 7)$

C. $(n - 5)(n + 7)$

D. $(n - 5)(n - 7)$ ☐

8. Factorize

(a) $x^2 + 8x + 7$

(2 marks)

(b) $y^2 - 11y - 26$

(2 marks)

9. (a) Factorize $x^2 - 2xy - 35y^2$

(2 marks)

(b) Hence, factorize $x^2 - 2xy - 35y^2 - 7x + 49y$

(3 marks)

10. Factorize

(a) $240uv - 72u^2 - 200v^2$

(3 marks)

(b) $63xy^2 + 28xz^2 + 84xyz$

(2 marks)

*11. Factorize $a^4 - a^2 - 2a^2b^2 - 2ab + b^4 - b^2$.

(7 marks)